



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-Section SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: Mini App – User Registration & Authentication

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1. Introduction

1.1. Purpose

The purpose of this document is to describe the design and system flow of a Mini Application that supports user registration, login, logout, and access to protected pages. This document is intended for students, instructors, and developers who will design, review, and implement the system. It serves as a reference for the system's functionality and structure before actual coding is done.

1.2. Scope

The system will allow users to create an account, log in, view a protected profile/dashboard, and log out. The system will restrict access to protected pages when the user is not authenticated. This documentation covers only the system design and diagrams (ERD, Use Case, Activity, Class, and Sequence diagrams). Actual coding and implementation are outside the scope of this document.

1.3. Definitions, Acronyms, and Abbreviations

- User – A person who uses the system
- Guest User – A user who is not logged in
- Authenticated User – A user who has successfully logged in
- ERD – Entity Relationship Diagram
- API – Application Programming Interface
- UI – User Interface
- JWT – JSON Web Token (used for authentication tokens)
- CRUD – Create, Read, Update, Delete

2. Overall Description

2.1. System Perspective

The system is a simple authentication module within a web application. It follows a client-server architecture where the frontend (React UI) communicates with a backend API (Spring Boot), which interacts with a database to store and retrieve user information. The authentication system may later be integrated into a larger application such as an e-commerce site, student portal, or management system.

2.2. User Classes and Characteristics

Guest User - Can access public pages and register or log in

Authenticated User - Can view profile/dashboard and access protected pages

2.3. Operating Environment

- Frontend: React
- Backend: Spring Boot (Java)
- Database: MySQL / PostgreSQL

- Tools:
 - IDE (IntelliJ / VS Code)
 - Browser (Chrome, Firefox)
 - API testing tool (Postman)
 - Platform: Web-based application running on a local or cloud server

2.4. Assumptions and Dependencies

Assumptions:

- Users have access to the internet and a modern web browser
- Users provide valid registration information
- Passwords are stored securely (hashed)

Dependencies:

- Database availability
- Backend API running properly
- Token provider or authentication library (e.g., JWT)
- Password encoding library

3. System Features and Functional Requirements

3.1. Feature 1: User Registration

Description: Allows a guest user to create a new account in the system by providing required details.

Functional Requirements:

- The system shall allow users to register using a username/email and password
- The system shall validate user input (e.g., required fields, password length)
- The system shall store user data securely in the database

3.2. Feature 2: User Login & Authentication

Description: Allows registered users to log in and gain access to protected features.

Functional Requirements:

- The system shall authenticate users using valid credentials
- The system shall generate an authentication token upon successful login
- The system shall restrict access to protected pages for unauthenticated users

3.3. Feature 3: View Profile/Dashboard

Description: Allows registered users to log in and gain access to protected features.

Functional Requirements:

- The system shall display user-specific information after login
- The system shall prevent guest users from accessing the dashboard
- The system shall retrieve user data from the database

3.4. Feature 2: User Logout

Description: Allows authenticated users to log out of the system and end their session.

Functional Requirements:

- The system shall invalidate the user session/token upon logout
- The system shall redirect the user to the login or home page
- The system shall prevent access to protected pages after logout

4. Non-Functional Requirements

Security:

- Passwords must be encrypted/hashed
- Authentication tokens must be securely generated and validated
- Protected routes must require authentication

Performance:

- Login and registration should respond within 2 seconds under normal load
- Usability:
- The UI should be simple and user-friendly
- Error messages should be clear and understandable

Reliability:

- The system should handle invalid login attempts gracefully
- The system should maintain session consistency

Scalability:

- The design should allow future expansion (e.g., roles, permissions, password reset)

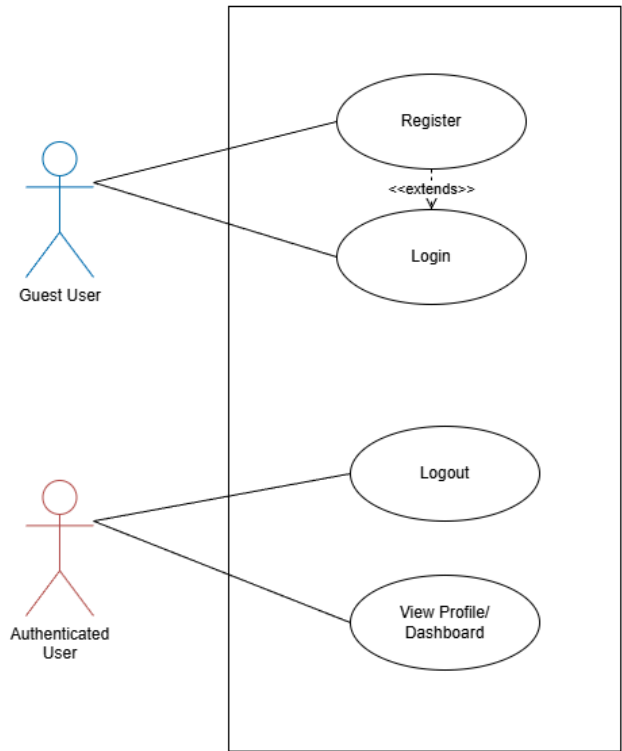
5. System Models (Diagrams)

5.1. ERD

USERS
user_id: BIGINT (PK)
username: VARCHAR(100) UNIQUE
email: VARCHAR(255) UNIQUE
password_hash: VARCHAR(255)
first_name: VARCHAR(50)
last_name: VARCHAR(50)
is_active: BOOLEAN
created_at: TIMESTAMP
updated_at: TIMESTAMP

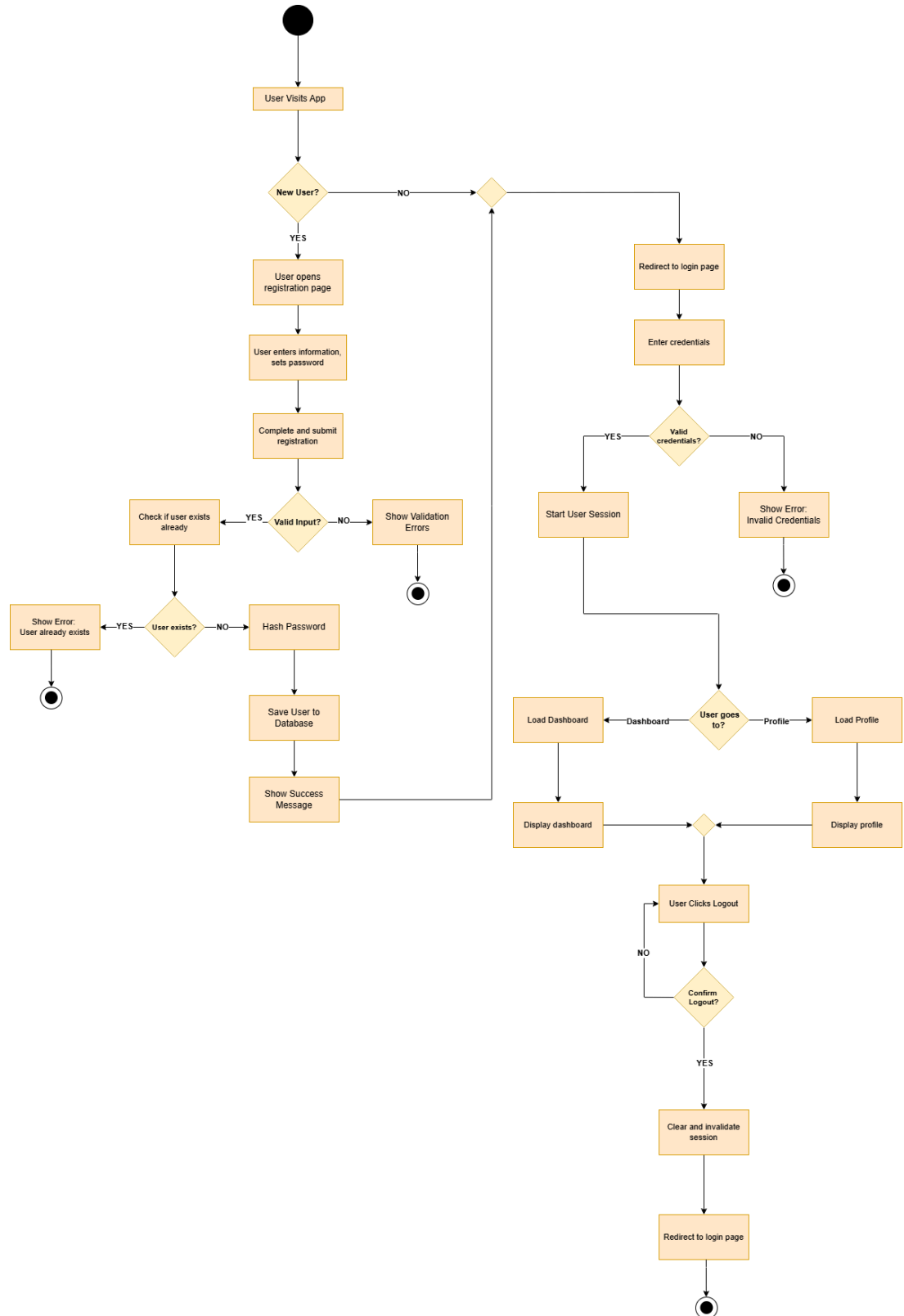
5.2. Use Case Diagram

Use Case Diagram



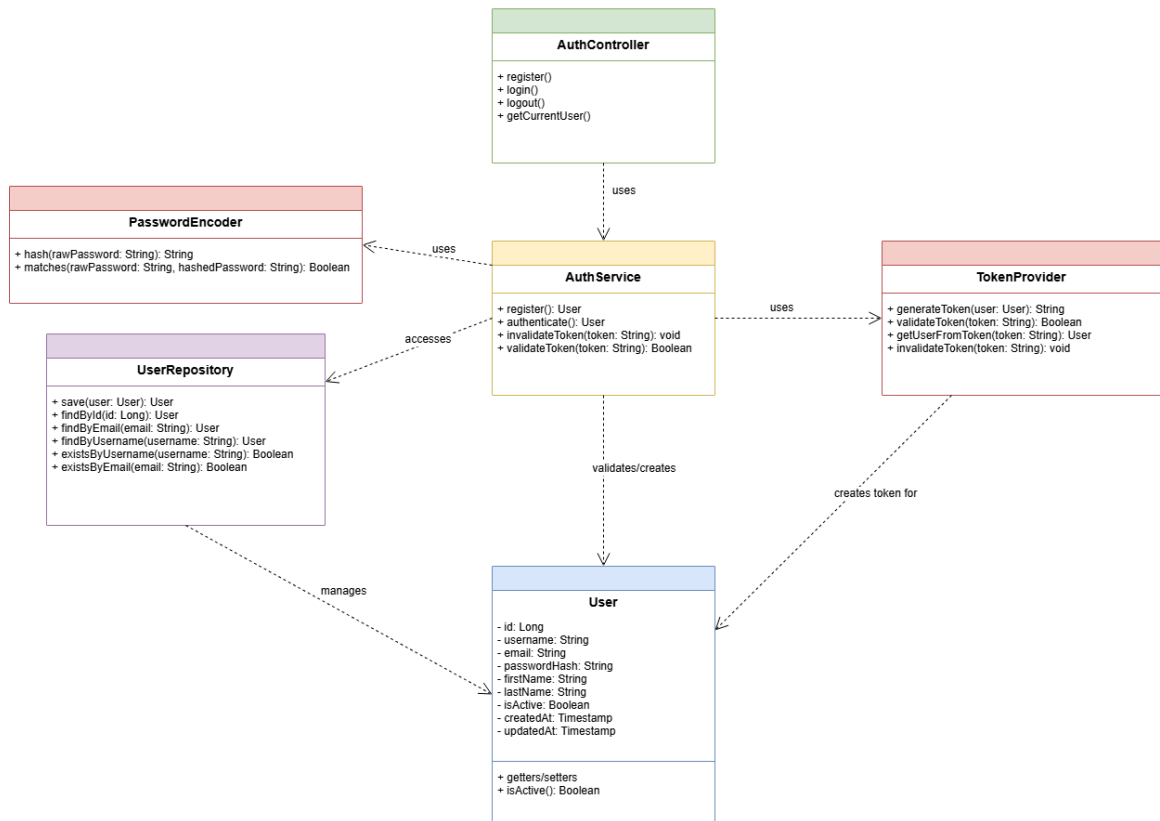
5.3. Activity Diagram

Activity Diagram - Registration, Login, and Logout Flow



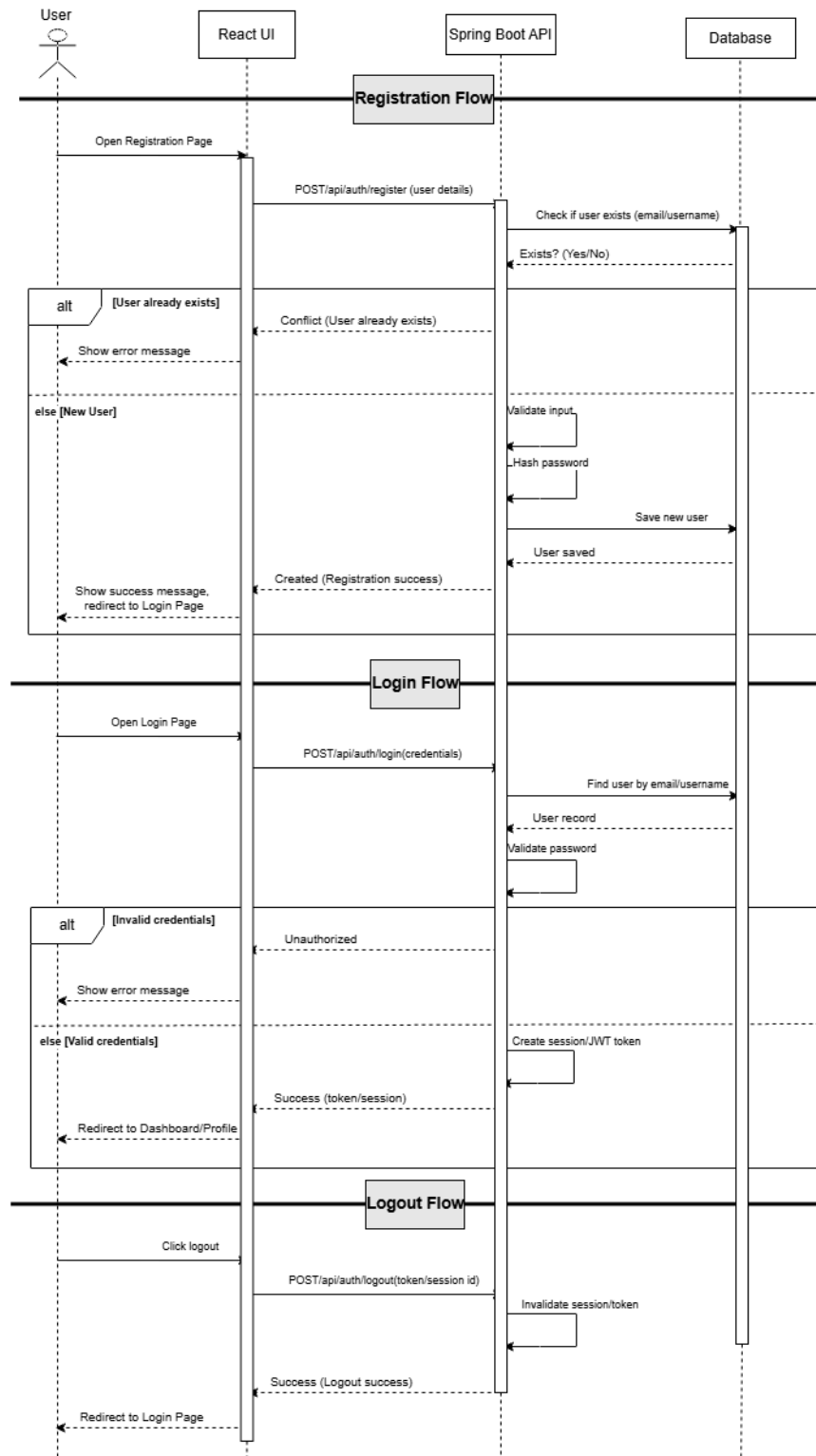
5.4. Class Diagram

Class Diagram - Authentication System Backend



5.5. Sequence Diagram

Mini App Register, Login, and Logout Sequence Diagram



6. Appendices

Activity Diagrams

GeeksforGeeks. (n.d.). UML | Activity Diagram. GeeksforGeeks. Retrieved February 8, 2026, from <https://www.geeksforgeeks.org/uml-activity-diagram/>

Sequence Diagrams

GeeksforGeeks. (n.d.). UML | Sequence Diagram. GeeksforGeeks. Retrieved February 8, 2026, from <https://www.geeksforgeeks.org/uml-sequence-diagram/>

Class Diagrams (Tutorial/Overview)

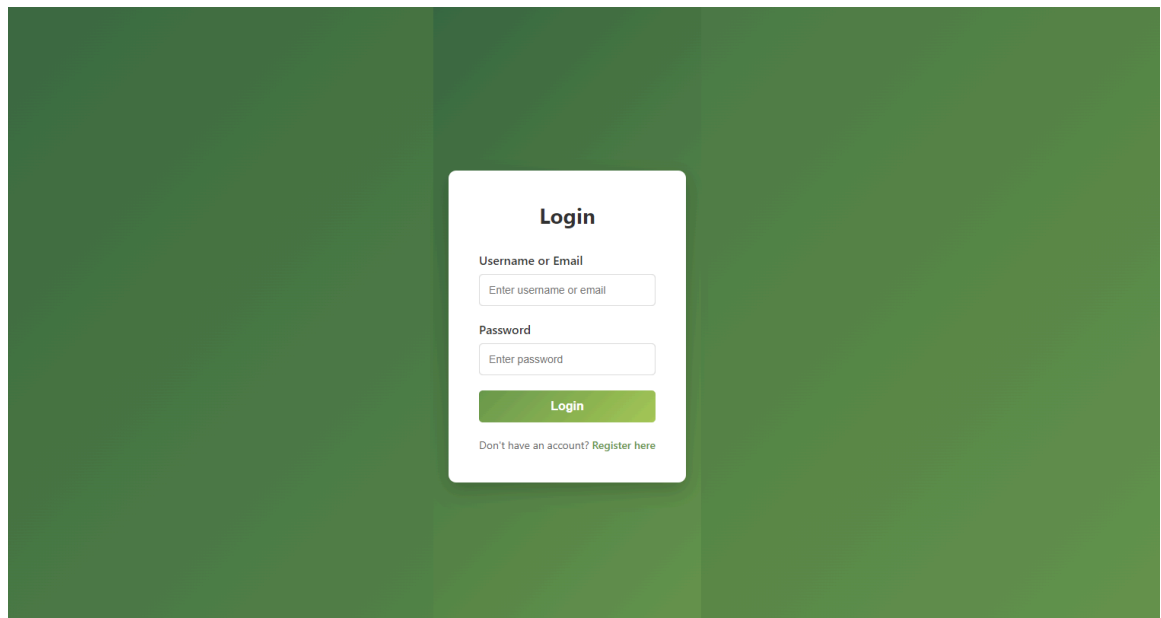
Fowler, M., & Scott, K. (2000). UML distilled: A brief guide to the standard object modeling language (3rd ed.). Addison-Wesley. (Chapter on Class Diagrams)

Class Diagrams (Online Reference)

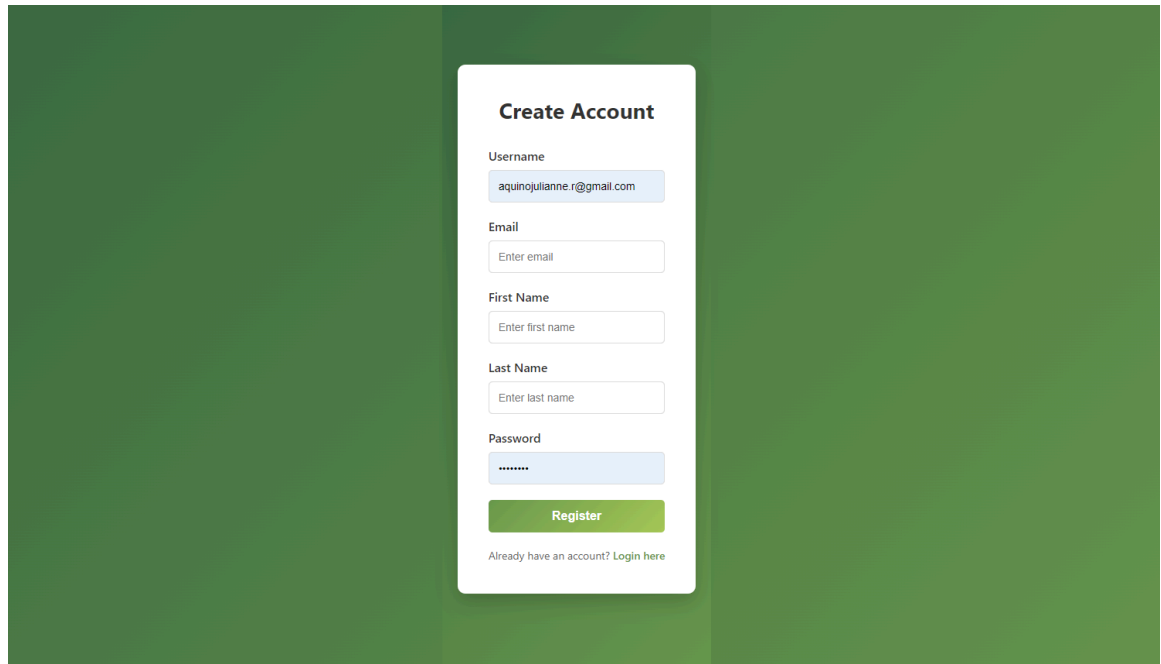
Sparx Systems. (n.d.). UML 2 Class Diagram. Sparx Systems UML Resources. Retrieved February 8, 2026, from https://sparxsystems.com/resources/uml2_tutorial/uml2_class_diagrams.html

7. Screenshots of the Web UI

a. Login Page (Index)



b. Register Page



The registration form is titled "Create Account" and is set against a dark green background. It contains several input fields: "Username" (pre-filled with "aquinojulianne.r@gmail.com"), "Email" (placeholder "Enter email"), "First Name" (placeholder "Enter first name"), "Last Name" (placeholder "Enter last name"), and "Password" (masked with "*****"). A green "Register" button is at the bottom, followed by a link "Already have an account? Login here".

Create Account

Username
aquinojulianne.r@gmail.com

Email
Enter email

First Name
Enter first name

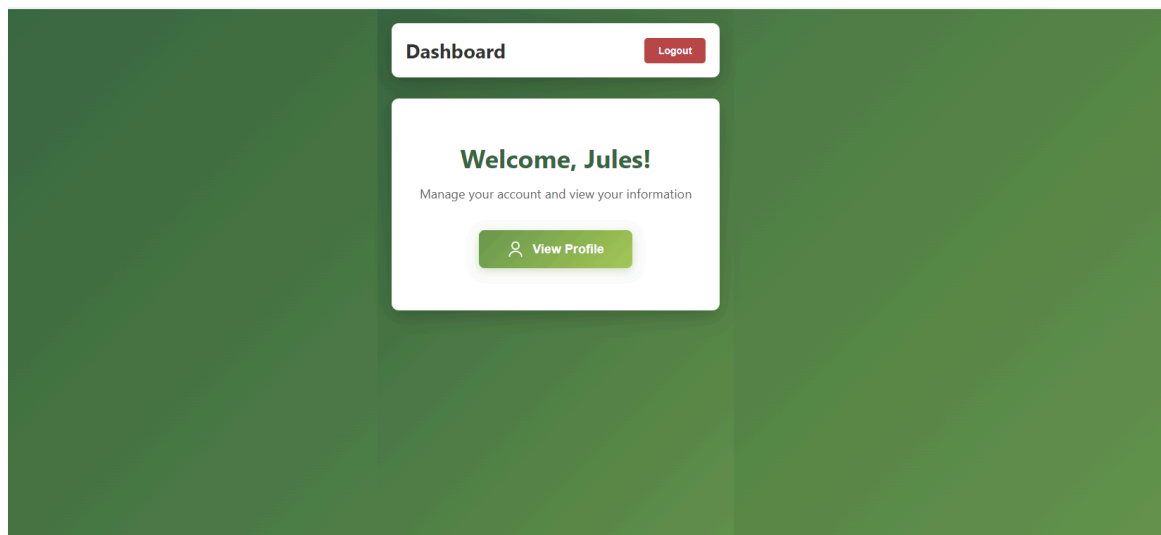
Last Name
Enter last name

Password

Register

[Already have an account? Login here](#)

c. Dashboard



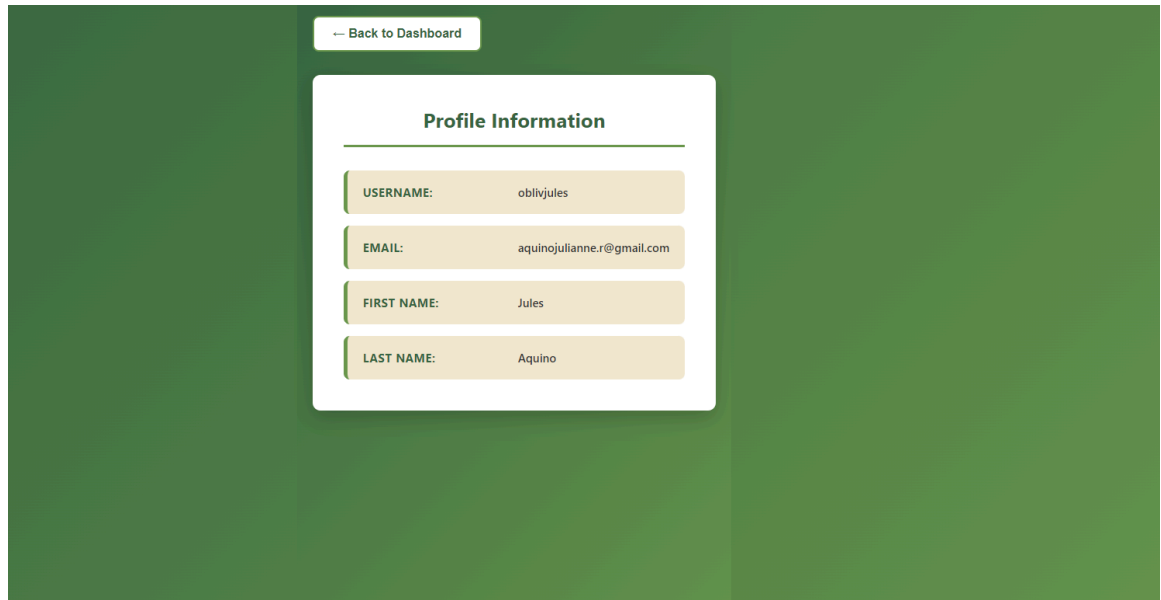
The dashboard features a "Dashboard" header with a "Logout" button. Below is a "Welcome, Jules!" message with the text "Manage your account and view your information" and a "View Profile" button with a user icon.

Dashboard **Logout**

Welcome, Jules!
Manage your account and view your information

 **View Profile**

d. Profile



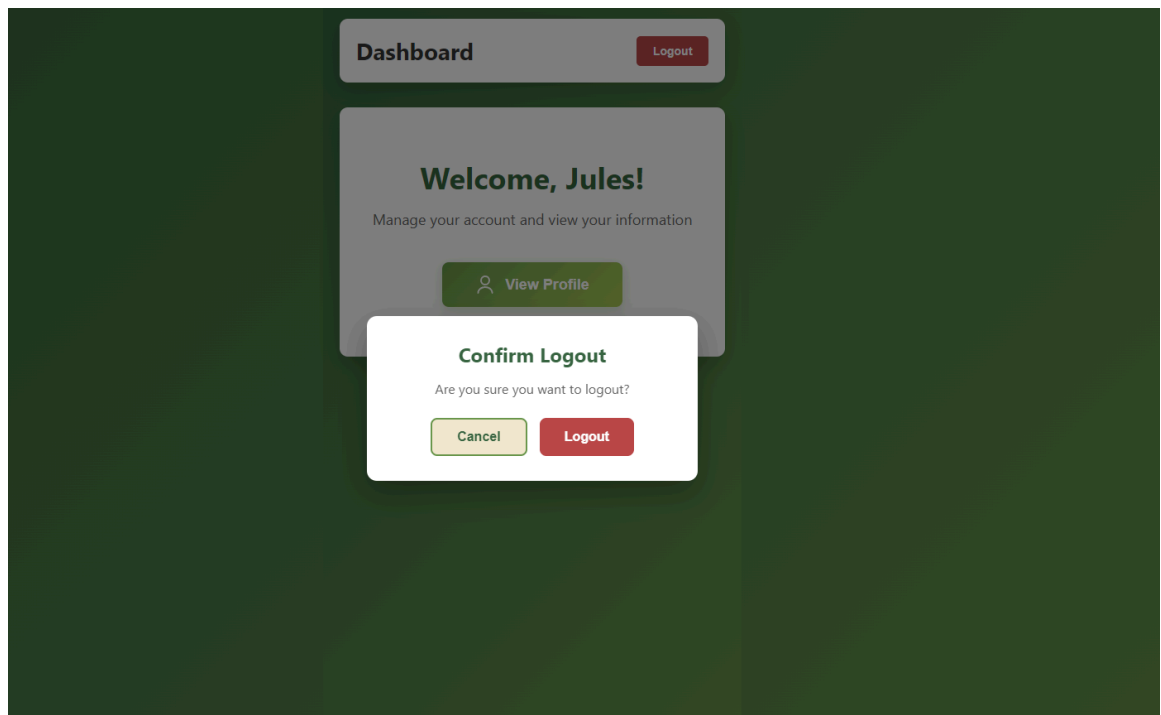
— Back to Dashboard

Profile Information

USERNAME:	oblivjules
EMAIL:	aquinojulianne.r@gmail.com
FIRST NAME:	Jules
LAST NAME:	Aquino

This screenshot shows a 'Profile Information' form on a green background. At the top left is a button labeled '— Back to Dashboard'. The form itself is a white card with a title 'Profile Information' and a horizontal line below it. It contains four rows of input fields, each with a label on the left and a value on the right: 'USERNAME: oblivjules', 'EMAIL: aquinojulianne.r@gmail.com', 'FIRST NAME: Jules', and 'LAST NAME: Aquino'.

e. Logout with Confirmation



Dashboard Logout

Welcome, Jules!

Manage your account and view your information

 View Profile

Confirm Logout

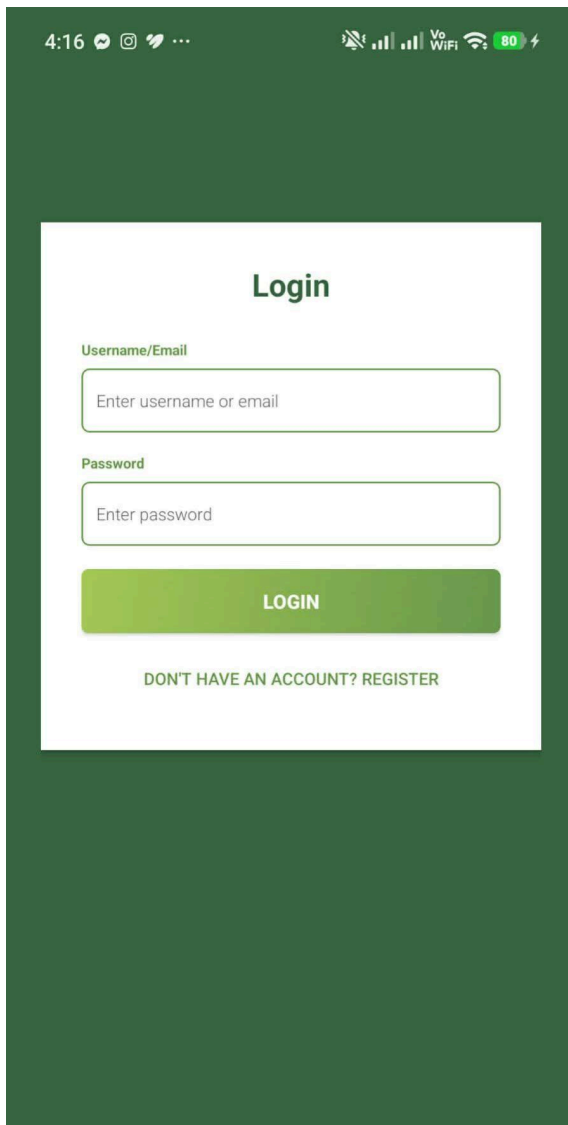
Are you sure you want to logout?

Cancel Logout

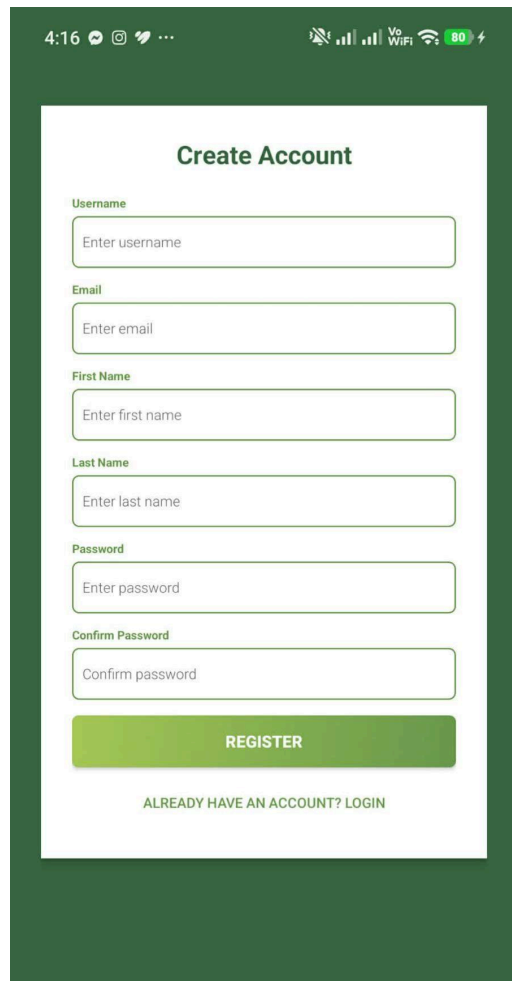
This screenshot shows a 'Dashboard' page with a dark green background. At the top, there's a header bar with 'Dashboard' on the left and a 'Logout' button on the right. Below this is a white card with the text 'Welcome, Jules!' and 'Manage your account and view your information'. A 'View Profile' button with a user icon is also present. A 'Confirm Logout' modal is open in the foreground, asking 'Are you sure you want to logout?' and providing 'Cancel' and 'Logout' buttons.

8. Screenshots of the Mobile UI

a. Login Page (Index)



b. Register Page



A mobile application registration screen with a dark green background. At the top, a status bar shows the time 4:16, social media icons, and connectivity status. The main content is a white card titled "Create Account". It contains six input fields: "Username" (placeholder: "Enter username"), "Email" (placeholder: "Enter email"), "First Name" (placeholder: "Enter first name"), "Last Name" (placeholder: "Enter last name"), "Password" (placeholder: "Enter password"), and "Confirm Password" (placeholder: "Confirm password"). Below the fields is a green "REGISTER" button. At the bottom of the card is a link that says "ALREADY HAVE AN ACCOUNT? LOGIN".

4:16

Create Account

Username

Enter username

Email

Enter email

First Name

Enter first name

Last Name

Enter last name

Password

Enter password

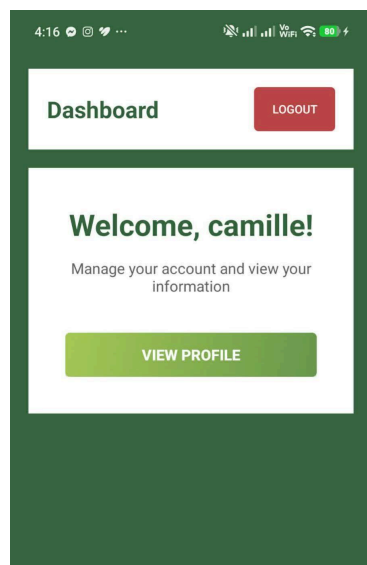
Confirm Password

Confirm password

REGISTER

ALREADY HAVE AN ACCOUNT? LOGIN

c. Dashboard



A mobile application dashboard screen with a dark green background. At the top, a status bar shows the time 4:16, social media icons, and connectivity status. The main content is a white card. At the top of the card is the word "Dashboard" and a red "LOGOUT" button. Below this is a large green "Welcome, camille!" message, followed by the text "Manage your account and view your information". At the bottom of the card is a green "VIEW PROFILE" button.

4:16

Dashboard **LOGOUT**

Welcome, camille!

Manage your account and view your information

VIEW PROFILE

d. Profile

A mobile app screenshot showing a 'Profile Information' screen. The screen has a dark green header with a '← BACK' button. The main content area is white and contains four stacked, light beige rectangular boxes. Each box has a label in bold green text and a value in black text. The status bar at the top shows the time as 4:16, signal strength, VoWiFi, and a battery level of 80%.

4:16 [icons] VoWiFi 80%

← BACK

Profile Information

USERNAME
camille

EMAIL
cam@gmail.com

FIRST NAME
camille

LAST NAME
aquino

e. Logout with Confirmation

A mobile app screenshot showing a 'Dashboard' screen. The screen has a dark green header with the word 'Dashboard' and a red 'LOGOUT' button. Below the header is a white card with the text 'Welcome, camille!' and 'Manage your account and view your information'. Below the card is a green 'VIEW PROFILE' button. A white dialog box is overlaid on the screen, titled 'Confirm Logout' with the text 'Are you sure you want to logout?'. The dialog has two buttons: 'CANCEL' and 'LOGOUT'. The status bar at the top shows the time as 4:16, signal strength, VoWiFi, and a battery level of 80%.

4:16 [icons] VoWiFi 80%

Dashboard LOGOUT

Welcome, camille!
Manage your account and view your information

VIEW PROFILE

Confirm Logout
Are you sure you want to logout?

CANCEL LOGOUT